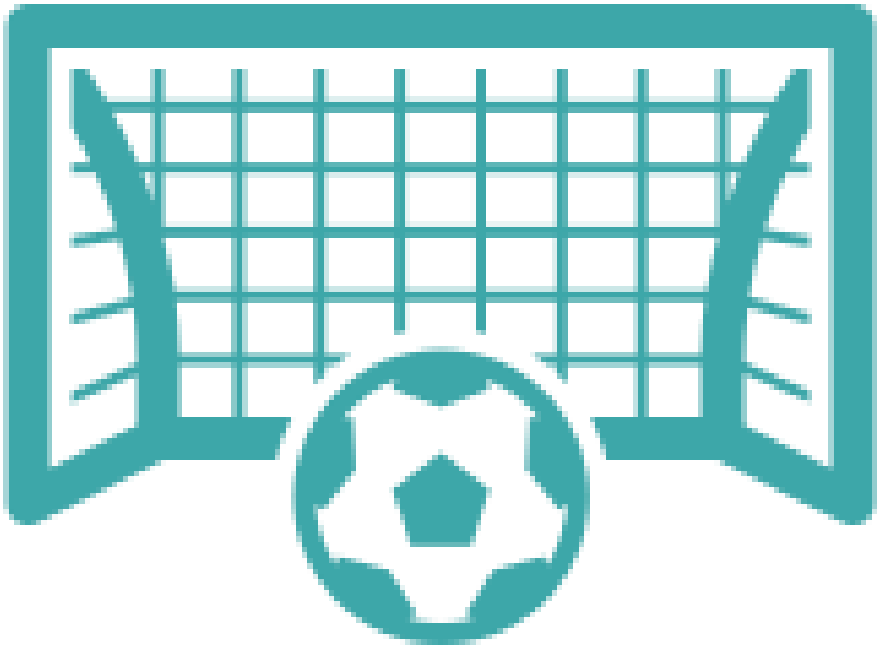
Football League

SQL: Database Design and Implementation



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Taline Manuelian

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Part 1: Database Design

1 – Normalisation Process

1.1 – UNF

The tables below are those shared in the brief in UNF rewritten up in Excel tables (Figures 1,2, and 3).

Consultant ID	Name	Team Name	Group Name
200	Aedan Petty	Data Masters	Cohort 4
201	Aliza Santos	Data Masters	Cohort 6
201	Kaylynn Vaughan	Data Masters	Training Team
202	Arjun Bauer	Data Masters	Cohort 7
202	Lilian Huber	Data Masters	Cohort 5
203	Lizeth Roberts	Data Masters	Training Team
203	Nathan Mcdowell	Data Masters	HR
204	Alvin Ali	Data Masters	Consultants
204	Jordin Christensen	Data Masters	HR
205	Saul Blevins	Data Masters	HR
205	Carina Meza	BI Gods	Cohort 5
206	Isabelle Campos	BI Gods	Bench
206	Kyleigh Phelps	BI Gods	Cohort 4
207	Angela Wong	BI Gods	HR
207	Kole Rojas	BI Gods	HR
208	Martha Potts	BI Gods	Cohort 7
208	Tomas Powell	BI Gods	Training Team
209	Paxton Clarke	BI Gods	Bench
209	Jamya Dodson	BI Gods	Consultants
210	Georgia Clements	BI Gods	Consultants
210	Edwin Crawford	Vis Wizards	Cohort 5
211	Malachi Osborn	Vis Wizards	Consultants
211	Zion Kent	Vis Wizards	Cohort 5
212	Anahi Reyes	Vis Wizards	Bench
212	Maddox Cabrera	Vis Wizards	Consultants
213	Brody Gutierrez	Vis Wizards	Cohort 7
213	Hayley Stevenson	Vis Wizards	Cohort 6
214	Kamora Sanchez	Vis Wizards	Consultants
214	Livia Holmes	Vis Wizards	Training Team
215	Tanner Jenkins	Vis Wizards	Consultants
215	Madelyn Meadows	Data Cleaners	Bench
216	Paola Wilkerson	Data Cleaners	Cohort 6
216	Jared Patton	Data Cleaners	Training Team
217	Pierre Washington	Data Cleaners	Consultants
217	Dominik Cochran	Data Cleaners	Cohort 7
218	Miya Skinner	Data Cleaners	Cohort 7
218	Mara Barnett	Data Cleaners	Bench
219	Cornelius Dodson	Data Cleaners	Cohort 5
219	Ashleigh Kaiser	Data Cleaners	Training Team
220	Weston Meza	Data Cleaners	Training Team

Figure 1 - Players table UNF

Date-Time	Home team	Away team	Venue	Week
01/10/2022 -- 16:00	Data Masters	BI Gods	Wimbeldon 1	1
01/10/2022 -- 16:00	Vis Wizards	Data Cleaners	Wimbeldon 2	1
08/10/2022 -- 16:00	Data Masters	Vis Wizards	Wimbeldon 2	2
08/10/2022 -- 16:00	BI Gods	Data Cleaners	Wimbeldon 3	2
22/10/2022 -- 16:00	Data Masters	Data Cleaners	Wimbeldon 3	3
22/10/2022 -- 16:00	BI Gods	Vis Wizards	Wimbeldon 2	3
29/10/2022 -- 16:00	BI Gods	Data Masters	Wimbeldon 1	4
29/10/2022 -- 16:00	Data Cleaners	Vis Wizards	Wimbeldon 3	4
05/11/2022 -- 16:00	Vis Wizards	Data Masters	Wimbeldon 3	5
05/11/2022 -- 16:00	Data Cleaners	BI Gods	Wimbeldon 1	5
12/11/2022 -- 16:00	Data Cleaners	Data Masters	Wimbeldon 2	6
12/11/2022 -- 16:00	Vis Wizards	BI Gods	Wimbeldon 3	6
12/11/2023 -- 16:00	Data Cleaners	Data Masters	Wimbeldon 1	Friendly
12/11/2023 -- 16:00	Vis Wizards	BI Gods	Wimbeldon 3	Friendly

Figure 2 - Fixtures table UNF

Date - Time	Home Team * Player(s) score	Score	Away Team * Player(s) score
01/10/2022 – 16:00	Data Masters * Alvin Ali (min 5) * Nathan Mcdowell (min 80)	2-0	BI Gods ** Isabelle Campos (min 10)
01/10/2022 – 16:00	Vis Wizards * Hayley Stevenson (min 11) ** Tanner Jenkins (min 30)	1 – 1	Data Cleaners * Dominik Cochran (min 10
08/10/2022 – 16:00	Data Masters * Alvin Ali (min 2) * Alvin Ali (min 88) ** Aedan Petty (min 10) ** Jordin Christensen (min 70)	2 – 1	Vis Wizards * Hayley Stevenson (min 85
08/10/2022 – 16:00	BI Gods * Jamya Dodson (min 20) * Tomas Powell (min 40)	2 – 1	Data Cleaners * Dominik Cochran (min 30) *** Paola Wilkerson (min 40
22/10/2022 – 16:00	Data Masters **Saul Blevins (min 35	0 – 0	Data Cleaners **Cornelius Dodson (min 88)
22/10/2022 – 16:00	BI Gods * Jamya Dodson (min 5)	1 – 2	Vis Wizards * Kamora Sanchez (min 2) * Hayley Stevenson (min 55)
29/10/2022 – 16:00	BI Gods * Jamya Dodson (min 1) * Martha Potts (min 82	2 – 2	Data Masters * Alvin Ali (min 35) * Kaylynn Vaughan (min 77) **Kaylynn Vaughan (min 60)
29/10/2022 – 16:00	Data Cleaners * Dominik Cochran (min 66) **Dominik Cochran (min 10	1 – 2	Vis Wizards * Maddox Cabrera (min 12) * Hayley Stevenson (min 68
05/11/2022 – 16:00	Vis Wizards * Livia Holmes (min 20) * Livia Holmes (min 25) * Brody Gutierrez (min 85) **Zion Kent (min 30)	3 – 2	Data Masters * Nathan Mcdowell (min 77) * Alvin Ali (min 89
05/11/2022 – 16:00	Data Cleaners ***Paola Wilkerson (min 10)	0 – 1	BI Gods * Jamya Dodson (min 75)
12/11/2022 – 16:00	Data Cleaners * Dominik Cochran (min 25) * Mara Barnett (min 46)	2-0	Data Masters **Aedan Petty (min 62)
12/11/2022 – 16:00	Vis Wizards *Brody Gutierrez (min 80) *Maddox Cabrera (min 90)	2 – 2	BI Gods *Jamya Dodson (min 29) *Paxton Clarke (min 87) **Tomas Powell (min 42
12/11/2023 - 16:00	Data Cleaners	To be Defined	Data Masters
12/11/2023 - 16:00	Vis Wizards	To be Defined	BI Gods

Figure 3 - Results table UNF

1.2 – 1NF

Many formatting errors were discovered in the UNF of the results table. Firstly, the scores and match events were all stored in one cell for each match. Each event was separated into its own row, and a primary key 'event_id' was added to distinguish each of the rows as its own event (Figure 4).

event_id	match_date	match_time	player_name	event_type	minute	side	team
1	01/10/2022	16:00	Alvin Ali	Goal	5	Home	Data Masters
2	01/10/2022	16:00	Nathan Mcdowell	Goal	80	Home	Data Masters
3	01/10/2022	16:00	Isabelle Campos	Yellow	10	Away	BI Gods
4	01/10/2022	16:00	Hayley Stevenson	Goal	11	Home	Vis Wizards
5	01/10/2022	16:00	Tanner Jenkins	Yellow	30	Home	Vis Wizards
6	01/10/2022	16:00	Dominik Cochran	Goal	10	Away	Data Cleaners
7	08/10/2022	16:00	Alvin Ali	Goal	2	Home	Data Masters
8	08/10/2022	16:00	Alvin Ali	Goal	88	Home	Data Masters
9	08/10/2022	16:00	Aedan Petty	Yellow	10	Home	Data Masters
10	08/10/2022	16:00	Jordin Christensen	Yellow	70	Home	Data Masters
11	08/10/2022	16:00	Hayley Stevenson	Goal	85	Away	Vis Wizards
12	08/10/2022	16:00	Jamya Dodson	Goal	20	Home	BI Gods
13	08/10/2022	16:00	Tomas Powell	Goal	40	Home	BI Gods
14	08/10/2022	16:00	Dominik Cochran	Goal	30	Away	Data Cleaners
15	08/10/2022	16:00	Paola Wilkerson	Red	40	Away	Data Cleaners
16	22/10/2022	16:00	Saul Blevins	Yellow	35	Home	Data Masters
17	22/10/2022	16:00	Cornelius Dodson	Yellow	88	Away	Data Cleaners
18	22/10/2022	16:00	Jamya Dodson	Goal	5	Home	BI Gods
19	22/10/2022	16:00	Kamora Sanchez	Goal	2	Away	Vis Wizards
20	22/10/2022	16:00	Hayley Stevenson	Goal	55	Away	Vis Wizards
21	29/10/2022	16:00	Jamya Dodson	Goal	1	Home	BI Gods
22	29/10/2022	16:00	Martha Potts	Goal	82	Home	BI Gods
23	29/10/2022	16:00	Alvin Ali	Goal	35	Away	Data Masters
24	29/10/2022	16:00	Kaylynn Vaughan	Goal	77	Away	Data Masters
25	29/10/2022	16:00	Kaylynn Vaughan	Yellow	60	Away	Data Masters
26	29/10/2022	16:00	Dominik Cochran	Goal	66	Home	Data Cleaners
27	29/10/2022	16:00	Dominik Cochran	Yellow	10	Home	Data Cleaners
28	29/10/2022	16:00	Maddox Cabrera	Goal	12	Away	Vis Wizards
29	29/10/2022	16:00	Hayley Stevenson	Goal	68	Away	Vis Wizards
30	05/11/2022	16:00	Livia Holmes	Goal	20	Home	Vis Wizards
31	05/11/2022	16:00	Livia Holmes	Goal	25	Home	Vis Wizards
32	05/11/2022	16:00	Brody Gutierrez	Goal	85	Home	Vis Wizards
33	05/11/2022	16:00	Zion Kent	Yellow	30	Home	Vis Wizards
34	05/11/2022	16:00	Nathan Mcdowell	Goal	77	Away	Data Masters
35	05/11/2022	16:00	Alvin Ali	Goal	89	Away	Data Masters
36	05/11/2022	16:00	Paola Wilkerson	Red	10	Home	Data Cleaners
37	05/11/2022	16:00	Jamya Dodson	Goal	75	Away	BI Gods
38	12/11/2022	16:00	Dominik Cochran	Goal	25	Home	Data Cleaners
39	12/11/2022	16:00	Mara Barnett	Goal	46	Home	Data Cleaners
40	12/11/2022	16:00	Aedan Petty	Yellow	62	Away	Data Masters
41	12/11/2022	16:00	Brody Gutierrez	Goal	80	Home	Vis Wizards
42	12/11/2022	16:00	Maddox Cabrera	Goal	90	Home	Vis Wizards
43	12/11/2022	16:00	Jamya Dodson	Goal	29	Away	BI Gods
44	12/11/2022	16:00	Paxton Clarke	Goal	87	Away	BI Gods
45	12/11/2022	16:00	Tomas Powell	Yellow	42	Away	BI Gods
	12/11/2023	16:00				Home	Data Cleaners
	12/11/2023	16:00				Away	Data Masters
	12/11/2023	16:00				Home	Vis Wizards
	12/11/2023	16:00				Away	BI Gods

Figure 4 - Results table 1NF, renamed to Events

The player table was found to have repeating values in the 'consultant_id' column, which is not atomic. For this reason, a primary key was added called 'player_id' to distinguish each consultant (Figure 5).

player_id (PK)	Consultant ID	player_name	team_name	group_name
1	200	Aedan Petty	Data Masters	Cohort 4
2	201	Aliza Santos	Data Masters	Cohort 6
3	201	Kaylynn Vaughan	Data Masters	Training Team
4	202	Arjun Bauer	Data Masters	Cohort 7
5	202	Lilian Huber	Data Masters	Cohort 5
6	203	Lizeth Roberts	Data Masters	Training Team
7	203	Nathan Mcdowell	Data Masters	HR
8	204	Alvin Ali	Data Masters	Consultants
9	204	Jordin Christensen	Data Masters	HR
10	205	Saul Blevins	Data Masters	HR
11	205	Carina Meza	BI Gods	Cohort 5
12	206	Isabelle Campos	BI Gods	Bench
13	206	Kyleigh Phelps	BI Gods	Cohort 4
14	207	Angela Wong	BI Gods	HR
15	207	Kole Rojas	BI Gods	HR
16	208	Martha Potts	BI Gods	Cohort 7
17	208	Tomas Powell	BI Gods	Training Team
18	209	Paxton Clarke	BI Gods	Bench
19	209	Jamya Dodson	BI Gods	Consultants
20	210	Georgia Clements	BI Gods	Consultants
21	210	Edwin Crawford	Vis Wizards	Cohort 5
22	211	Malachi Osborn	Vis Wizards	Consultants
23	211	Zion Kent	Vis Wizards	Cohort 5
24	212	Anahi Reyes	Vis Wizards	Bench
25	212	Maddox Cabrera	Vis Wizards	Consultants
26	213	Brody Gutierrez	Vis Wizards	Cohort 7
27	213	Hayley Stevenson	Vis Wizards	Cohort 6
28	214	Kamora Sanchez	Vis Wizards	Consultants
29	214	Livia Holmes	Vis Wizards	Training Team
30	215	Tanner Jenkins	Vis Wizards	Consultants
31	215	Madelyn Meadows	Data Cleaners	Bench
32	216	Paola Wilkerson	Data Cleaners	Cohort 6
33	216	Jared Patton	Data Cleaners	Training Team
34	217	Pierre Washington	Data Cleaners	Consultants
35	217	Dominik Cochran	Data Cleaners	Cohort 7
36	218	Miya Skinner	Data Cleaners	Cohort 7
37	218	Mara Barnett	Data Cleaners	Bench
38	219	Cornelius Dodson	Data Cleaners	Cohort 5
39	219	Ashleigh Kaiser	Data Cleaners	Training Team
40	220	Weston Meza	Data Cleaners	Training Team

Figure 5 - Players table 1NF

As for the fixtures, a column to distinguish between repeating dates was not evident, so the column 'match_id' was created as the primary key to separate each match clearly. 'Match_type' was also established to show which matches are seasonal or friendly games. Additionally, the week numbers for friendly games were left blank as they aren't part of the league (Figure 6).

match_id (PK)	match_date	Home team	Away team	Venue	week_number	match_time	match_type
1	01/10/2022	Data Masters	BI Gods	Wimbeldon 1	1	16:00	Season
2	01/10/2022	Vis Wizards	Data Cleaners	Wimbeldon 2	1	16:00	Season
3	08/10/2022	Data Masters	Vis Wizards	Wimbeldon 2	2	16:00	Season
4	08/10/2022	BI Gods	Data Cleaners	Wimbeldon 3	2	16:00	Season
5	22/10/2022	Data Masters	Data Cleaners	Wimbeldon 3	3	16:00	Season
6	22/10/2022	BI Gods	Vis Wizards	Wimbeldon 2	3	16:00	Season
7	29/10/2022	BI Gods	Data Masters	Wimbeldon 1	4	16:00	Season
8	29/10/2022	Data Cleaners	Vis Wizards	Wimbeldon 3	4	16:00	Season
9	05/11/2022	Vis Wizards	Data Masters	Wimbeldon 3	5	16:00	Season
10	05/11/2022	Data Cleaners	BI Gods	Wimbeldon 1	5	16:00	Season
11	12/11/2022	Data Cleaners	Data Masters	Wimbeldon 2	6	16:00	Season
12	12/11/2022	Vis Wizards	BI Gods	Wimbeldon 3	6	16:00	Season
13	12/11/2023	Data Cleaners	Data Masters	Wimbeldon 1		16:00	Friendly
14	12/11/2023	Vis Wizards	BI Gods	Wimbeldon 3		16:00	Friendly

Figure 6 - Fixtures table 1NF, renamed to Match

1.3 – 2NF

The 1NF of the players table is also in 2NF as at this point, the match tables are properly separated.

To ensure the events table is in 2NF, match_id was added as a foreign key from the fixtures table as well as player_id. This meant match_date and match_time could be referenced from the fixtures table, and player_name and team_name could be referenced from the players table. Now the events table's non-key attributes (event_type, minute) depend on the whole key (event_id), not partially on match/player (Figure 7).

event_id (PK)	match_id (FK)	player_id (FK)	event_type	minute
1	1	8	Goal	5
2	1	7	Goal	80
3	1	12	Yellow	10
4	2	27	Goal	11
5	2	30	Yellow	30
6	2	35	Goal	10
7	3	8	Goal	2
8	3	8	Goal	88
9	3	1	Yellow	10
10	3	9	Yellow	70
11	3	27	Goal	85
12	4	19	Goal	20
13	4	17	Goal	40
14	4	35	Goal	30
15	4	32	Red	40
16	5	10	Yellow	35
17	5	38	Yellow	88
18	6	19	Goal	5
19	6	28	Goal	2
20	6	27	Goal	55
21	7	19	Goal	1
22	7	16	Goal	82
23	7	8	Goal	35
24	7	3	Goal	77
25	7	3	Yellow	60
26	8	35	Goal	66
27	8	35	Yellow	10
28	8	25	Goal	12
29	8	27	Goal	68
30	9	29	Goal	20
31	9	29	Goal	25
32	9	26	Goal	85
33	9	23	Yellow	30
34	9	7	Goal	77
35	9	8	Goal	89
36	10	32	Red	10
37	10	19	Goal	75
38	11	35	Goal	25
39	11	37	Goal	46
40	11	1	Yellow	62
41	12	26	Goal	80
42	12	25	Goal	90
43	12	19	Goal	29
44	12	18	Goal	87
45	12	17	Yellow	42

Figure 7 - Event table 2NF and 3NF

1.4 – 3NF

Issues identified at this stage in the 3NF for the players table was that team_name and group_name have repeating values in their columns, which cause update anomalies. Therefore, lookup tables were created, and the team and group names were replaced with their ids, acting as foreign keys (Figures 8, 9, and 10).

player_id (PK)	consultant_id	player_name	team_id (FK)	group_id (FK)
1	200	Aedan Petty	1	1
2	201	Aliza Santos	1	3
3	201	Kaylynn Vaughan	1	6
4	202	Arjun Bauer	1	3
5	202	Lilian Huber	1	2
6	203	Lizeth Roberts	1	6
7	203	Nathan Mcdowell	1	7
8	204	Alvin Ali	1	8
9	204	Jordin Christensen	1	7
10	205	Saul Blevins	1	7
11	205	Carina Meza	2	2
12	206	Isabelle Campos	2	5
13	206	Kyleigh Phelps	2	1
14	207	Angela Wong	2	7
15	207	Kole Rojas	2	7
16	208	Martha Potts	2	4
17	208	Tomas Powell	2	6
18	209	Paxton Clarke	2	5
19	209	Jamya Dodson	2	8
20	210	Georgia Clements	2	8
21	210	Edwin Crawford	3	2
22	211	Malachi Osborn	3	8
23	211	Zion Kent	3	2
24	212	Anahi Reyes	3	5
25	212	Maddox Cabrera	3	8
26	213	Brody Gutierrez	3	4
27	213	Hayley Stevenson	3	3
28	214	Kamora Sanchez	3	8
29	214	Livia Holmes	3	6
30	215	Tanner Jenkins	3	8
31	215	Madelyn Meadows	4	5
32	216	Paola Wilkerson	4	3
33	216	Jared Patton	4	6
34	217	Pierre Washington	4	8
35	217	Dominik Cochran	4	4
36	218	Miya Skinner	4	4
37	218	Mara Barnett	4	5
38	219	Cornelius Dodson	4	2
39	219	Ashleigh Kaiser	4	6
40	220	Weston Meza	4	6

Figure 8 - Players table 3NF

team_id (PK)	team_name
1	Data Masters
2	BI Gods
3	Vis Wizards
4	Data Cleaners

Figure 9 - Team table 3NF

group_id (PK)	group_name
1	Cohort 4
2	Cohort 5
3	Cohort 6
4	Cohort 7
5	Bench
6	Training Team
7	HR
8	Consultants

Figure 10 - Group table 3NF

The fixtures table had repeating values in both the venue and team name columns. To solve this, a separate lookup table was created to store the venue names with a new primary key of the venue_id, and that venue_id was inserted as a foreign key into the fixtures table (Figure 11).

venue_id (PK)	venue_name
1	Wimbeldon 1
2	Wimbeldon 2
3	Wimbeldon 3

Figure 11 - Venue table 3NF

Also, the team_id was added as a foreign key to replace the team_name but is separated to showcase which is the home and away team for each match (Figure 12).

match_id (PK)	match_date	match_time	home_team_id (FK)	away_team_id (FK)	venue_id (FK)	week_number	match_type
1	01/10/2022	16:00	1	2	1	1	Season
2	01/10/2022	16:00	3	4	2	1	Season
3	08/10/2022	16:00	1	3	2	2	Season
4	08/10/2022	16:00	2	4	3	2	Season
5	22/10/2022	16:00	1	4	3	3	Season
6	22/10/2022	16:00	2	3	2	3	Season
7	29/10/2022	16:00	2	1	1	4	Season
8	29/10/2022	16:00	4	3	3	4	Season
9	05/11/2022	16:00	3	1	3	5	Season
10	05/11/2022	16:00	4	2	1	5	Season
11	12/11/2022	16:00	4	1	2	6	Season
12	12/11/2022	16:00	3	2	3	6	Season
13	12/11/2023	16:00	4	1	1		Friendly
14	12/11/2023	16:00	3	2	3		Friendly

Figure 12 - Match table 3NF

As for the events table, the 2NF was already in 3NF so it remains the same, as all the names are stored under one team, group, and venue, and all other tables reference them via IDs (Figure 7).

1.5 – Changes to 3NF

Based on feedback from Task 1, changes were implemented to only store 'goals' under 'event_type' in the table 'EVENT' and was therefore removed under 'MATCH' to avoid any inconsistencies when it came to querying the data as part of Task 2. These changes were reflected in Figure 13 through the addition of 'RESULT' to store points.

result_id (PK)	points	match_id (FK)	team_id (FK)
1	3	1	1
2	1	1	2
3	0	2	3

Figure 13 - Result table 3NF

The final seven tables in 3NF are: Team, Group, Player, Venue, Match, Event, and Result.

2 – ER Diagram: Updates to the Design

Figure 14 shows the original ER Diagram designed as part of Task 1.

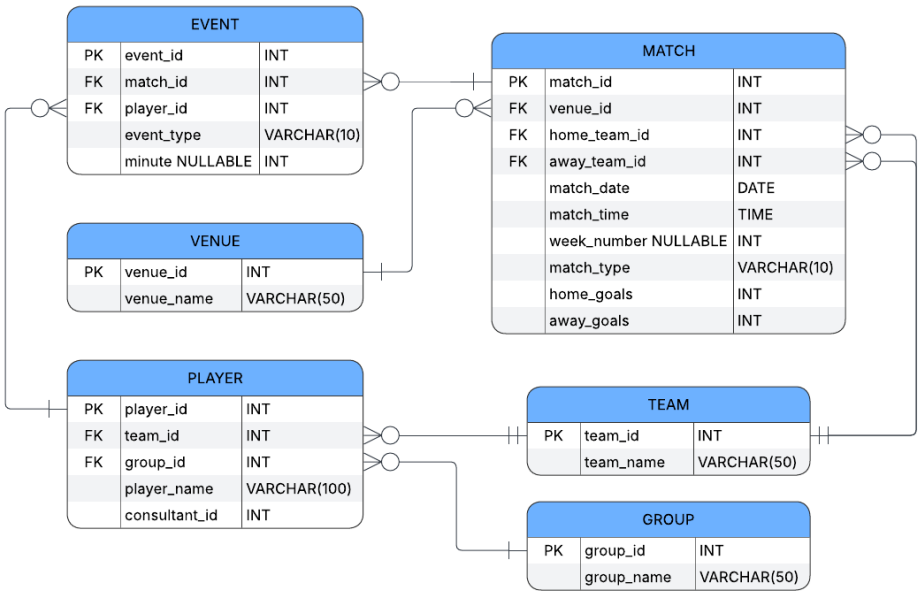

Figure 14 - ER Diagram Crow's feet, UML notation [Task 1 Original]

Figure 15 shows the updated ER Diagram based on changes made to the normalisation process mentioned in section 1.5.

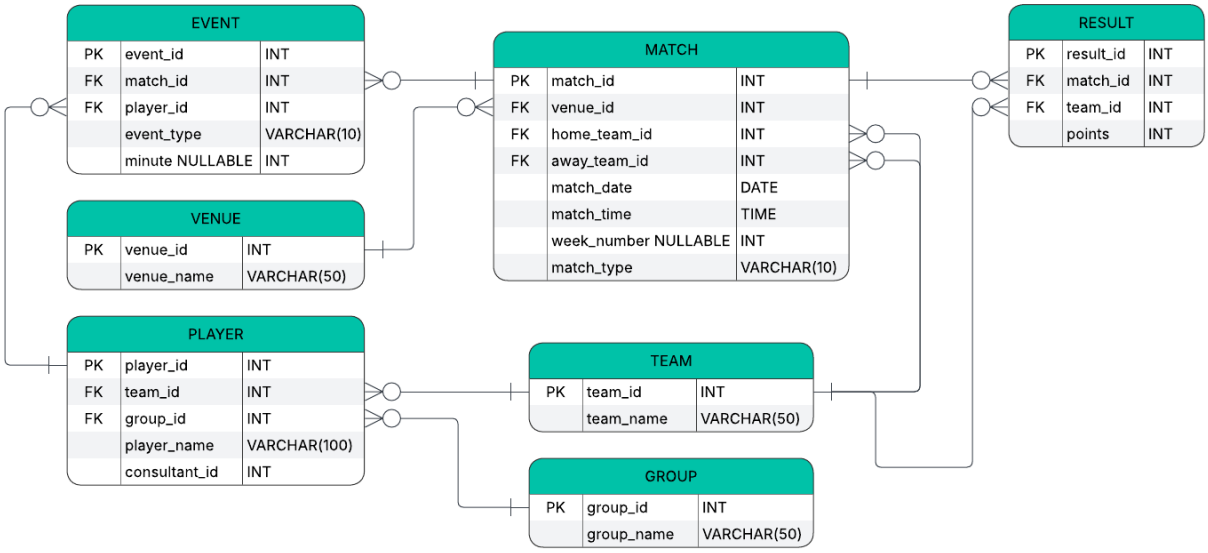

Figure 15 - ER Diagram Crow's feet, UML notation [Task 2 Updated]

3 – Data Dictionary: Updated

The following data dictionary describes the relational schema designed for the 6-a-side Company Football League database. The design is in Third Normal Form (3NF). All primary keys (PKs) and foreign keys (FKs) are explicitly defined, and integrity constraints are specified to ensure data consistency. All tables in 3NF can be viewed in the Excel sheet here.

The normalisation process was adjusted and re-done to reflect the same changes implemented in the adjusted ER Diagram. All tables remain the same except 'Match'

no longer stores home_goals and away_goals in order to avoid any inconsistencies in the data, and there is an addition of a new table 'Result' to store the points gained by each team after each match, which can be referred to under *section 3.7**.*

All columns are NOT NULL unless NULLABLE is noted.

3.1 - TEAM

'Team' stores information about participating teams. Team names are stored once to eliminate redundancy and prevent update anomalies.

KEY	FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
PK	*team_id (UNIQUE)*	INT	Integer		Unique identifier for each team	1
	team_name (UNIQUE)	VARCHAR	Text	50	Team name (club name). Should be unique	Data Masters

3.2 - GROUP

'Group' stores participating groups players are in (8 in total).

KEY	FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
PK	*group_id (UNIQUE)*	INT	Integer		Unique identifier for each participating group	7
	group_name (UNIQUE)	VARCHAR	Text	50	Group name. Should be unique	HR

3.3 - PLAYER

'Player' stores consultants participating in the football tournament.

KEY	FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
PK	*player_id (UNIQUE)*	INT	Integer		Unique identifier for a player record	1
FK	*team_id*	INT	Integer		References TEAM(team_id) which is the team the player belongs to	1
FK	*group_id*	INT	Integer		References GROUP(group_id) which is the group the player belongs to	1
	player_name	VARCHAR	Text	100	Player's full name	Aeden Petty

	consultant_id	INT	Integer		Consultant identifier from the source dataset (stored for traceability). Not a unique value	200

3.4 – VENUE

'Venue' stores available football pitches.

KEY	***FIELD NAME***	***DATA TYPE***	***DATA FORMAT***	***FIELD SIZE***	***DESCRIPTION***	***EXAMPLE***
PK	*venue_id (UNIQUE)*	INT	Integer		Unique identifier for each venue the match is held at	1
	venue_name (UNIQUE)	VARCHAR	Text	50	Name of the venue	Wimbledon 1

3.5 – MATCH

'Match' stores both season and friendly matches.

KEY	***FIELD NAME***	***DATA TYPE***	***DATA FORMAT***	***FIELD SIZE***	***DESCRIPTION***	***EXAMPLE***
PK	*match_id (UNIQUE)*	INT	Integer		Unique identifier for each match fixture	1
FK	*venue_id*	INT	Integer		References VENUE(venue_id) which is the venue where the match is played	1
FK	*home_team_id*	INT	Integer		References TEAM(team_id) which is the home team for the match	1
FK	*away_team_id*	INT	Integer		References TEAM(team_id) which is the away team for the match	1
	match_date	DATE	DD-MM-YYYY		Date the match is played	01/10/2022
	match_time	TIME	HH:MM		Time the match begins	16:00
	week_number (NULLABLE)	INT	Integer		League week number for seasonal	1

					matches, NULL for friendly matches	
	match_type	VARCHAR	Text	10	Indicates whether the match is part of the season or a friendly. Allowed values: Season, Friendly	Season

3.6 - EVENT

'Event' stores match incidents such as goals and disciplinary actions.

KEY	***FIELD NAME***	***DATA TYPE***	***DATA FORMAT***	***FIELD SIZE***	***DESCRIPTION***	***EXAMPLE***
PK	*event_id (UNIQUE)*	INT	Integer		Unique identifier for each match event	1
FK	*match_id*	INT	Integer		References MATCH(match_id) which is the unique identifier for each match fixture	1
FK	*player_id*	INT	Integer		References PLAYER(player_id) which is the unique identifier the player the event is associated with	1
	event_type	VARCHAR	Text	10	Type of event recorded in the match such as goals or cards given: GOAL, YELLOW, RED. Additional could be added within the field size	GOAL
	minute	INT	Integer		Minute when the event occurred during the match. Must be between 0 and 90 as there is no extra time normally in 6-a-side football matches	1

3.7 – RESULT

'Result' stores the points gained by each team from each match to indicate whether they won, there was a draw, or they lost. This table stores the points outcome per team per match, so there are typically two rows per match where one row is the home team and the other the away team.

KEY	FIELD NAME	DATA TYPE	DATA FORMAT	FIELD SIZE	DESCRIPTION	EXAMPLE
PK	*result_id (UNIQUE)*	INT	Integer		Unique identifier for each result outcome after each match for each team	1
FK	*match_id*	INT	Integer		References MATCH(match_id) which is the unique identifier for the match in which the result occurred in	1
FK	*team_id*	INT	Integer		References TEAM(team_id) which is the unique identifier for the team the result has occurred for	1
	points (NULLABLE)	INT	Integer		Points awarded for a specific match to a specific team. Must be 0, 1, or 3 (0 indicates loss, 1 indicates draw, or 3 indicates win). NULL for future matches.	3

4 – Assumptions

The following assumptions were made during the design of the 6-a-side Company Football League database according to each table created:

4.1 – TEAM

1. *The league season consists of four teams*
2. *Each team will play against each other team twice within one tournament*

4.2 – GROUP

3. *There exist 8 participating groups in which a player can be placed into*

4.3 – PLAYER

4. *Each player (consultant) belongs to exactly one participating group*
5. *Each player may play for any team*
6. *Each player may play for only one team*
7. *Each player has a consultant ID which is a non-unique identifier*

4.4 – VENUE

8. *Three venues are available for scheduling matches: Wimbledon 1, Wimbledon 2, Wimbledon 3*
9. *The availability of venues will be assigned according to the fixtures at the beginning of the tournament*

4.5 – MATCH

10. *Each match involves exactly two teams: one home team and one away team*
11. *A match is played at a single venue*
12. *Matches can be either league or friendly matches*
13. *League matches are organised into weeks*
14. *Friendly matches are not part of the season and therefore may not have a week number*
15. *Friendly matches are supported by the system*

4.6 – EVENT

16. *Match events (goals and cards given) are recorded individually per player*
17. *As this is 6-a-side football, extra time is not normally played*

4.7 - RESULT

18. *The scoring system follows standard rules: 3 points for a win, 1 point for a draw, and 0 points for a loss*
19. *Friendly scores are stored in the system but aren't part of the final league standings*
20. *The database stores future matches that are scheduled to occur, so matches that haven't occurred yet have points stored as 'NULL'*

5 – Constraints

5.1 – Entity Integrity

- *All primary keys (PK) are unique and NOT NULL*
- *Each entity has a clearly defined PK:*
    - *TEAM(team_id)*
    - *GROUP(group_id)*
    - *VENUE(venue_id)*
    - *PLAYER(player_id)*
    - *MATCH(match_id)*
    - *EVENT(event_id)*
    - *RESULT(result_id)*

5.2 – Referential Integrity

The following foreign key relationships are enforced:

- *PLAYER.team_id → TEAM(team_id)*
- *PLAYER.group_id → GROUP(group_id)*
- *MATCH.home_team_id → TEAM(team_id)*
- *MATCH.away_team_id → TEAM(team_id)*
- *MATCH.venue_id → VENUE(venue_id)*
- *EVENT.match_id → MATCH(match_id)*
- *EVENT.player_id → PLAYER(player_id)*
- *RESULT.match_id → MATCH(match_id)*
- *RESULT.team_id → TEAM(team_id)*

5.3 – Business Constraints

5.3.1 – PLAYER

- *Each player belongs to exactly one team*
- *Each player belongs to exactly one group*

5.3.2 – MATCH

- *Each match must have exactly one home team and one away team: home_team_id <> away_team_id (a team cannot play itself)*
- *home_team_id ≠ away_team_id*
- *Friendly handling:*
  - *match_type can either be 'Season' or 'Friendly'*
  - *week_number is NULL for Friendly matches*
  - *If match_type = 'Season' then week_number IS NOT NULL*
  - *Only matches with match_type = 'Season' are included in league standings*

5.3.3 – EVENT

- *event_type can either be 'Goal', 'Yellow', 'Red', or any other event if it were to be added (within the character limit)*
- *minute must be between 0 and 90*
- *Events are optional per match (a match may have zero events)*
- *Events must always reference a valid match and a valid player*
- *Goals are derived from EVENT records where event_type = 'GOAL': scoreline for a match is derived by counting GOAL events for players belonging to each team*

5.3.4 – RESULT

- *points can either be 0, 1, or 3 where 0 is a loss, 1 a draw, and 3 a win*
- *Each match must have exactly two RESULT rows: one for the home team and one for the away team (crow's foot can't show "exactly 2" well)*
- *Only matches with match_type = 'Season' contribute to league table calculations (friendlies can still have RESULT rows, but must be excluded in standings queries)*

Part 2: Implementation & Testing

As part of the process of implementing the database design, the DDL was created to define the database structures in tables. DML commands were then used to manipulate the data stored in the database tables. New records were inserted from the sample data provided in the brief and from our final structured 3NF tables in Excel. Once all data was stored, the database implementation was tested using sample test to ensure it operated correctly and could be used for future queries.

1 – Scripts

1.1 – DDL

1.1.1 – DDL Issues Identified

One main issue found during the DDL creation process was the naming of tables and columns. The table GROUP had to be renamed to "group" so that it was in quotation marks, as it was getting confused with the group function in the software (Figures 16 and 17).

```
24  -- 2) GROUP
25  CREATE TABLE group (
26    group_id INT PRIMARY KEY,
27    group_name VARCHAR(50) NOT NULL UNIQUE
28  );
29
30  -- 3) VENUE
31  CREATE TABLE venue (
```

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```
ERROR:  syntax error at or near "group"
LINE 2: CREATE TABLE group (
                     ^

SQL state: 42601
Character: 26
```

Figure 16 - 'Group' table naming error identified

```sql
-- 2) GROUP (group needs to be quoted as it is a reserved keyword)
CREATE TABLE "group" (
  group_id INT PRIMARY KEY,
  group_name VARCHAR(50) NOT NULL UNIQUE
);
```

Figure 17- Correction of 'Group' table, renaming to '"group"'

Although there was no error, another table that had to be renamed was the table MATCH, as it was also being identified by the software as a function (Figure 18).

```sql
-- 5) MATCH
CREATE TABLE match (
  match_id INT PRIMARY KEY,
  match_date DATE NOT NULL,
  match_time TIME NOT NULL,
  venue_id INT NOT NULL,
  home_team_id INT NOT NULL,
  away_team_id INT NOT NULL,
  week_number INT NULL,
  match_type VARCHAR(10) NOT NULL,
```

Figure 18 - 'Match' table naming difference identified

This table was therefore renamed to MATCH_FIXTURE to distinguish the two (Figure 19).

```sql
-- 5) MATCH
CREATE TABLE match_fixture (-- match renamed to match_fixture as could be confusing with function 'match')
  match_id INT PRIMARY KEY,
  match_date DATE NOT NULL,
  match_time TIME NOT NULL,
  venue_id INT NOT NULL,
  home_team_id INT NOT NULL,
  away_team_id INT NOT NULL,
  week_number INT NULL,
  match_type VARCHAR(10) NOT NULL,
```

Figure 19 - Correcting 'Match' table, renaming to 'Match_fixture'

'Minute' also had to be renamed to 'game_minute' to ensure clarity and distinction between functions and column/table names (Figures 20 and 21).

```sql
-- 6) EVENT (goals/cards stored here)
CREATE TABLE event (
  event_id INT PRIMARY KEY,
  match_id INT NOT NULL,
  player_id INT NOT NULL,
  event_type VARCHAR(10) NOT NULL,
  minute INT NOT NULL,
```

Figure 20 - 'Minute' column naming difference identified

```
-- 6) EVENT (goals/cards stored here)
CREATE TABLE event (
  event_id INT PRIMARY KEY,
  match_id INT NOT NULL,
  player_id INT NOT NULL,
  event_type VARCHAR(10) NOT NULL,
  game_minute INT NOT NULL, -- minute renamed to game_minute
```

Figure 21 - Correcting 'Minute' column, renaming to 'game_minute'

1.1.2 – DDL Implementation

After all issues were identified and corrected, the full DDL could be created. This can be viewed below:

```
/* FOOTBALL LEAGUE DATABASE
      DDL: TABLE CREATION   */

-- reset for during development
DROP TABLE IF EXISTS result CASCADE;
DROP TABLE IF EXISTS event CASCADE;
DROP TABLE IF EXISTS match_fixture CASCADE;
DROP TABLE IF EXISTS player CASCADE;
DROP TABLE IF EXISTS venue CASCADE;
DROP TABLE IF EXISTS team CASCADE;
DROP TABLE IF EXISTS "group" CASCADE;

-- Set DateStyle to DD/MM/YYYY
SET datestyle = 'DMY';

-- 1) TEAM
CREATE TABLE team (
  team_id INT PRIMARY KEY,
  team_name VARCHAR(50) NOT NULL UNIQUE
);

SELECT *
FROM team;

-- 2) GROUP (group needs to be quoted as it is a reserved keyword)
CREATE TABLE "group" (
  group_id INT PRIMARY KEY,
  group_name VARCHAR(50) NOT NULL UNIQUE
);

-- 3) VENUE
CREATE TABLE venue (
  venue_id INT PRIMARY KEY,
```

```
  venue_name VARCHAR(50) NOT NULL UNIQUE
);

-- 4) PLAYER
CREATE TABLE player (
  player_id INT PRIMARY KEY,
  consultant_id INT NOT NULL,
  player_name VARCHAR(100) NOT NULL UNIQUE,
  team_id INT NOT NULL,
  group_id INT NOT NULL,

  -- foreign keys to player
  CONSTRAINT fk_player_team FOREIGN KEY (team_id) REFERENCES team(team_id),
  CONSTRAINT fk_player_group FOREIGN KEY (group_id) REFERENCES
"group"(group_id)
);

-- 5) MATCH
CREATE TABLE match_fixture (-- match renamed to match_fixture as could be
confusing with function 'match')
  match_id INT PRIMARY KEY,
  match_date DATE NOT NULL,
  match_time TIME NOT NULL,
  venue_id INT NOT NULL,
  home_team_id INT NOT NULL,
  away_team_id INT NOT NULL,
  week_number INT NULL,
  match_type VARCHAR(10) NOT NULL,

  -- foreign keys to match
  CONSTRAINT fk_match_venue FOREIGN KEY (venue_id) REFERENCES venue(venue_id),
  CONSTRAINT fk_match_home_team FOREIGN KEY (home_team_id) REFERENCES
team(team_id),
  CONSTRAINT fk_match_away_team FOREIGN KEY (away_team_id) REFERENCES
team(team_id),

  -- Domain constraints
  CONSTRAINT chk_match_type CHECK (match_type IN ('Season','Friendly')),
  CONSTRAINT chk_teams_different CHECK (home_team_id <> away_team_id),

  -- Friendly vs Season week rule
  CONSTRAINT chk_week_by_type CHECK (
    (match_type = 'Friendly' AND week_number IS NULL)
    OR (match_type = 'Season' AND week_number IS NOT NULL)
  ));
```

```
-- 6) EVENT (goals/cards stored here)
CREATE TABLE event (
  event_id INT PRIMARY KEY,
  match_id INT NOT NULL,
  player_id INT NOT NULL,
  event_type VARCHAR(10) NOT NULL,
  game_minute INT NOT NULL, -- minute renamed to game_minute

  -- foreign keys to event
  CONSTRAINT fk_event_match FOREIGN KEY (match_id) REFERENCES
match_fixture(match_id),
  CONSTRAINT fk_event_player FOREIGN KEY (player_id) REFERENCES
player(player_id),

  -- event range of options (can be ammended if there are new additions later
on, for now only these)
  CONSTRAINT chk_event_type CHECK (event_type IN ('Goal','Yellow','Red')),
  CONSTRAINT chk_game_minute CHECK (game_minute BETWEEN 0 AND 90)
  );

-- 7) RESULT: stores points per team per match
CREATE TABLE result (
  result_id INT PRIMARY KEY,
  match_id INT NOT NULL,
  team_id INT NOT NULL,
  points INT NULL, --future games points will be NULL as they are yet to occur

  -- constraint that each match must have exactly two result rows: one for
home and one for away team
  CONSTRAINT uq_result_match_team UNIQUE (match_id, team_id),

  -- foreign keys to result
  CONSTRAINT fk_result_match FOREIGN KEY (match_id) REFERENCES
match_fixture(match_id),
  CONSTRAINT fk_result_team FOREIGN KEY (team_id) REFERENCES team(team_id),

  -- constraint for point options
  CONSTRAINT chk_points CHECK (points IN (0,1,3))
);
```

1.2 – DML

1.2.1 – DML Issues Identified

A key issue that was discovered was the existence of a different date format in the system of 'MM/DD/YYYY' which was recognising days as months instead of vice versa. Using the function below this was solved and corrected to ensure the dates are all read in the form 'DD/MM/YYYY' (Figure 22).

```
-- Set DateStyle to DD/MM/YYYY
SET datestyle = 'DMY';
```

Figure 22 - Adjusting DateStyle to DD/MM/YYYY

This piece of code was placed at the beginning of the script to ensure clarity with the data settings for the whole project.

1.2.2 – DML Implementation

Each table was populated with the data provided in the scenario:

```
/* DML: REFERENCE DATA */

-- 1) TEAM
INSERT INTO team (team_id, team_name) VALUES
(1,'Data Masters'),
(2,'BI Gods'),
(3,'Vis Wizards'),
(4,'Data Cleaners');

-- 2) GROUP
INSERT INTO "group" (group_id, group_name) VALUES
(1,'Cohort 4'),
(2,'Cohort 5'),
(3,'Cohort 6'),
(4,'Cohort 7'),
(5,'Bench'),
(6,'Training Team'),
(7,'HR'),
(8,'Consultants');

-- 3) VENUE
INSERT INTO venue (venue_id, venue_name) VALUES
(1,'Wimbeldon 1'),
(2,'Wimbeldon 2'),
(3,'Wimbeldon 3');

/* DML: TRANSACTIONAL DATA */

-- 4) Players
INSERT INTO player (player_id,
consultant_id,
```

```
player_name,
team_id,
group_id) VALUES
(1,200,'Aedan Petty',1,1),
(2,201,'Aliza Santos',1,3),
(3,202,'Kalynn Vaughan',1,6),
(4,203,'Arjun Bauer',1,3),
(5, 202,'Lilian Huber',1,2),
(6,    203,'Lizeth Roberts',1,6),
(7,    203,'Nathan Mcdowell',1,7),
(8,    204,'Alvin Ali',1,8),
(9,    204,'Jordin Christensen',1,7),
(10, 205,'Saul Blevins',1,7),
(11, 205,'Carina Meza',2,2),
(12, 206,'Isabelle Campos',2,5),
(13, 206,'Kyleigh Phelps',2,1),
(14, 207,'Angela Wong',2,7),
(15, 207,'Kole Rojas',2,7),
(16, 208,'Martha Potts',2,4),
(17, 208,'Tomas Powell',2,6),
(18, 209,'Paxton Clarke',2,5),
(19, 209,'Jamya Dodson',2,8),
(20, 210,'Georgia Clements',2,8),
(21, 210,'Edwin Crawford',3,2),
(22, 211,'Malachi Osborn',3,8),
(23, 211,'Zion Kent',3,2),
(24, 212,'Anahi Reyes',3,5),
(25, 212,'Maddox Cabrera',3,8),
(26, 213,'Brody Gutierrez',3,4),
(27, 213,'Hayley Stevenson',3,3),
(28, 214,'Kamora Sanchez',3,8),
(29, 214,'Livia Holmes',3,6),
(30, 215,'Tanner Jenkins',3,8),
(31, 215,'Madelyn Meadows',4,5),
(32, 216,'Paola Wilkerson',4,3),
(33, 216,'Jared Patton',4,6),
(34, 217,'Pierre Washington',4,8),
(35, 217,'Dominik Cochran',4,4),
(36, 218,'Miya Skinner',4,4),
(37, 218,'Mara Barnett',4,5),
(38, 219,'Cornelius Dodson',4,2),
(39, 219,'Ashleigh Kaiser',4,6),
(40, 220,'Weston Meza',4,6)
;

-- 5) Matches
```

```
INSERT INTO match_fixture (match_id,
match_date,
match_time,
venue_id,
home_team_id,
away_team_id,
week_number,
match_type) VALUES
(1,'01/10/2022','16:00',1,1,     2,     1,     'Season'),
(2,'01/10/2022','16:00',2,3,     4,     1,     'Season'),
(3,'08/10/2022','16:00',2,1,     3,     2,     'Season'),
(4,'08/10/2022','16:00',3,2,     4,     2, 'Season'),
(5,'22/10/2022','16:00',3,1,     4,     3,     'Season'),
(6,'22/10/2022','16:00',2,2,     3,     3,     'Season'),
(7,'29/10/2022','16:00',1,2,     1,     4,     'Season'),
(8,'29/10/2022','16:00',3,4,     3,     4,     'Season'),
(9,'05/11/2022','16:00',3,3,     1,     5,     'Season'),
(10,'05/11/2022','16:00',1,      4,     2,     5,     'Season'),
(11,'12/11/2022','16:00',2,      4,     1,     6,     'Season'),
(12,'12/11/2022','16:00',3,      3,     2,     6, 'Season'),
(13,'12/11/2023','16:00',1,      4,     1,NULL,'Friendly'),
(14,'12/11/2023','16:00',3,      3,     2,NULL,      'Friendly')
;

-- 6) Events
INSERT INTO event (event_id,
match_id,
player_id,
event_type,
game_minute) VALUES
(1,    1,     8,     'Goal',      5),
(2, 1, 7,     'Goal',      80),
(3,    1,     12,    'Yellow',10),
(4,    2,     27,    'Goal', 11),
(5,    2,     30,    'Yellow',30),
(6,    2,     35,    'Goal',      10),
(7,    3,     8,     'Goal',      2),
(8,    3,     8,     'Goal',      88),
(9,    3,     1,     'Yellow',10),
(10, 3,       9,     'Yellow',70),
(11,3, 27,    'Goal',      85),
(12,4, 19,    'Goal',      20),
(13,4, 17,    'Goal',      40),
(14,4, 35,    'Goal',      30),
(15,4, 32,    'Red', 40),
(16,5, 10,    'Yellow',35),
```

```
(17,5, 38, 'Yellow',88),
(18,6, 19,    'Goal',      5),
(19,6, 28,    'Goal',      2),
(20,6, 27,    'Goal',      55),
(21,7, 19,    'Goal',      1),
(22,7, 16,    'Goal',      82),
(23,7, 8,     'Goal',      35),
(24,7, 3,     'Goal',      77),
(25,7, 3,     'Yellow',60),
(26,8, 35,    'Goal',      66),
(27,8, 35,    'Yellow',10),
(28,8, 25,    'Goal',      12),
(29,8, 27,    'Goal',      68),
(30,9, 29,    'Goal',      20),
(31,9, 29,    'Goal',      25),
(32,9, 26,    'Goal',      85),
(33,9, 23,    'Yellow',30),
(34,9, 7,     'Goal',      77),
(35,9, 8,     'Goal',      89),
(36,10,       32,    'Red', 10),
(37,10,       19,    'Goal',      75),
(38,11,       35,    'Goal',      25),
(39,11,       37,    'Goal',      46),
(40,11, 1,    'Yellow',62),
(41,12, 26,   'Goal',      80),
(42,12,       25,    'Goal',      90),
(43,12,       19,    'Goal',      29),
(44,12,       18, 'Goal',  87),
(45,12,       17,    'Yellow',42)
;

-- 7) Results: TWO rows per match (one per team)
-- Example: match 1 home team got 3, away team got 1
INSERT INTO result (result_id,
match_id,
team_id,
points) VALUES
(1,1,1,3),
(2,1,2,0),
(3,2,3,1),
(4,2,4,1),
(5,3,1,3),
(6,3,3,0),
(7,4,2,3),
(8,4,4,0),
(9,5,1,1),
```

```
(10,5,4,1),
(11,6,2,0),
(12,6,3,3),
(13,7,2,1),
(14,7,1,1),
(15,8,4,0),
(16,8,3,3),
(17,9,3,3),
(18,9,1,0),
(19,10,4,0),
(20,10,2,3),
(21,11,4,3),
(22,11,1,0),
(23,12,3,1),
(24,12,2,1),
(25,13,4,NULL),
(26,13,1,NULL),
(27,14,3,NULL),
(28,14,2,NULL)
;
```

2 – Testing

To ensure the DDL and DML implementation was successful, the data was then queried using 7 sample questions as a form of testing the script generated. No issues were identified during the testing stage as all our DDL and DML issues were corrected before moving on to this stage. The queries and their outputs can be seen below under each subsection.

2.1 – Question 1

```
/* TESTING: QUERYING THE DATA */

/* 1) Listing all students, who play for a particular department
(i.e. Cohort 4 group). */
SELECT p.player_id, p.player_name, t.team_name, g.group_name
FROM player p
JOIN team t ON t.team_id = p.team_id
JOIN "group" g ON g.group_id = p.group_id
WHERE g.group_name = 'Cohort 4'
ORDER BY p.player_name;
```

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	player_id integer	player_name character varying (100)	team_name character varying (50)	group_name character varying (50)
1	1	Aedan Petty	Data Masters	Cohort 4
2	13	Kyleigh Phelps	BI Gods	Cohort 4

Figure 23 - Question 1 output

2.2 – Question 2

```
/* 2) Listing all fixtures for a specific date
(i.e. 29th of October 2022 and including team names and venues). */
SELECT
  m.match_id,
  m.match_date,
  m.match_time,
  ht.team_name AS home_team,
  at.team_name AS away_team,
  v.venue_name,
  m.match_type,
  m.week_number
FROM match_fixture m
JOIN team ht ON ht.team_id = m.home_team_id
JOIN team at ON at.team_id = m.away_team_id
JOIN venue v ON v.venue_id = m.venue_id
WHERE m.match_date = DATE '29-10-2022'
ORDER BY m.match_time, m.match_id;
```

Data Output Messages Notifications

	match_id integer	match_date date	match_time time without time zone	home_team character varying (50)	away_team character varying (50)	venue_name character varying (50)	match_type character varying (10)	week_number integer
1	7	2022-10-29	16:00:00	BI Gods	Data Masters	Wimbeldon 1	Season	4
2	8	2022-10-29	16:00:00	Data Cleaners	Vis Wizards	Wimbeldon 3	Season	4

Figure 24 - Question 2 output

2.3 – Question 3

```
-- 3) Listing all the players who have scored more than 2 goals.
SELECT
  p.player_id,
  p.player_name,
  t.team_name,
  COUNT(*) AS goals_scored
FROM event e
JOIN player p ON p.player_id = e.player_id
```

```
JOIN team t   ON t.team_id = p.team_id
WHERE e.event_type = 'Goal'
GROUP BY p.player_id, p.player_name, t.team_name
HAVING COUNT(*) > 2
ORDER BY goals_scored DESC, p.player_name;
```

Data Output Messages Notifications

	player_id integer	player_name character varying (100)	team_name character varying (50)	goals_scored bigint
1	8	Alvin Ali	Data Masters	5
2	19	Jamya Dodson	BI Gods	5
3	35	Dominik Cochran	Data Cleaners	4
4	27	Hayley Stevenson	Vis Wizards	4

Figure 25 - Question 3 output

2.4 – Question 4

```
-- 4) Listing the number of cards (yellow and red) per team.
SELECT
  t.team_name,
  SUM(CASE WHEN e.event_type = 'Yellow' THEN 1 ELSE 0 END) AS yellow_cards,
  SUM(CASE WHEN e.event_type = 'Red' THEN 1 ELSE 0 END) AS red_cards,
  COUNT(*) AS total_cards
FROM event e
JOIN player p ON p.player_id = e.player_id
JOIN team t ON t.team_id = p.team_id
WHERE e.event_type IN ('Yellow', 'Red')
GROUP BY t.team_name
ORDER BY total_cards DESC, t.team_name;
```

Data Output Messages Notifications

	team_name character varying (50)	yellow_cards bigint	red_cards bigint	total_cards bigint
1	Data Masters	5	0	5
2	Data Cleaners	2	2	4
3	BI Gods	2	0	2
4	Vis Wizards	2	0	2

Figure 26 - Question 4 output

2.5 - Question 5

```
-- 5) Return the games that are going to be played (friendly matches).
SELECT
  m.match_type,
  m.match_id,
  m.match_date,
  m.match_time,
  ht.team_name AS home_team,
  at.team_name AS away_team,
  v.venue_name
FROM match_fixture m
JOIN team ht ON ht.team_id = m.home_team_id
JOIN team at ON at.team_id = m.away_team_id
JOIN venue v ON v.venue_id = m.venue_id
WHERE m.match_type = 'Friendly'
ORDER BY m.match_date, m.match_time;
```

Data Output Messages Notifications

	match_type character varying (10)	match_id integer	match_date date	match_time time without time zone	home_team character varying (50)	away_team character varying (50)	venue_name character varying (50)
1	Friendly	13	2023-11-12	16:00:00	Data Cleaners	Data Masters	Wimbeldon 1
2	Friendly	14	2023-11-12	16:00:00	Vis Wizards	BI Gods	Wimbeldon 3

Figure 27 - Question 5 output

2.6 – Question 6

```
-- 6) The table displays the team's name and the points earned during the
tournament.
SELECT
  t.team_name,
  COALESCE(SUM(r.points), 0) AS points
```

```
FROM team t
LEFT JOIN result r ON r.team_id = t.team_id
LEFT JOIN match_fixture m ON m.match_id = r.match_id
WHERE (m.match_type = 'Season' OR m.match_type IS NULL) -- keeps teams with
zero matches
GROUP BY t.team_name
ORDER BY points DESC, t.team_name;
```

Data Output Messages Notifications

	team_name character varying (50)	points bigint
1	Vis Wizards	11
2	BI Gods	8
3	Data Masters	8
4	Data Cleaners	5

Figure 28 - Question 6 output

2.7 – Question 7

```
/* 7) For each team, present the distribution of goals scored
and conceded in each half of the match. */
SELECT t.team_name,
    -- Goals Scored: First Half
    COUNT(CASE
            WHEN e.event_type = 'Goal'
             AND e.game_minute BETWEEN 0 AND 45
             AND p.team_id = t.team_id
            THEN 1
        END) AS goals_scored_first_half,

      -- Goals Scored: Second Half
      COUNT(CASE
            WHEN e.event_type = 'Goal'
             AND e.game_minute BETWEEN 46 AND 90
             AND p.team_id = t.team_id
            THEN 1
        END) AS goals_scored_second_half,

      -- Goals Scored: Full Match Totals
```

```
    COUNT(CASE
            WHEN e.event_type = 'Goal'
             AND p.team_id = t.team_id
            THEN 1
         END) AS goals_scored_full_match,

       -- Goals Conceded: First Half
    COUNT(CASE
            WHEN e.event_type = 'Goal'
             AND e.game_minute BETWEEN 0 AND 45
             AND (
                  (p.team_id = m.home_team_id AND t.team_id = m.away_team_id)
               OR (p.team_id = m.away_team_id AND t.team_id = m.home_team_id)
                 )
            THEN 1
         END) AS goals_conceded_first_half,

    -- Goals Conceded: Second Half
    COUNT(CASE
            WHEN e.event_type = 'Goal'
             AND e.game_minute BETWEEN 46 AND 90
             AND (
                  (p.team_id = m.home_team_id AND t.team_id = m.away_team_id)
               OR (p.team_id = m.away_team_id AND t.team_id = m.home_team_id)
                 )
            THEN 1
         END) AS goals_conceded_second_half,

       -- Goals Conceded: Full Match Totals
    COUNT(CASE
            WHEN e.event_type = 'Goal'
             AND (
                  (p.team_id = m.home_team_id AND t.team_id = m.away_team_id)
               OR (p.team_id = m.away_team_id AND t.team_id = m.home_team_id)
                 )
            THEN 1
         END) AS goals_conceded_full_match

FROM team t
JOIN match_fixture m
    ON t.team_id IN (m.home_team_id, m.away_team_id)
JOIN event e
    ON e.match_id = m.match_id
JOIN player p
    ON p.player_id = e.player_id
```

```
GROUP BY t.team_name
ORDER BY goals_scored_full_match DESC, t.team_name;
```

Data Output Messages Notifications

Showing rows: 1 to 4

	team_name character varying (50)	goals_scored_first_half bigint	goals_scored_second_half bigint	goals_scored_full_match bigint	goals_conceded_first_half bigint	goals_conceded_second_half bigint	goals_conceded_full_match bigint
1	Vis Wizards	5	6	11	4	5	9
2	BI Gods	5	3	8	4	5	9
3	Data Masters	3	5	8	4	4	8
4	Data Cleaners	3	2	5	4	2	6

Figure 29 - Question 7 output

3 – PostgreSQL ER Diagram

In PostgreSQL, an ERD was generated based on the DDL created which can be seen below in Figure 30.

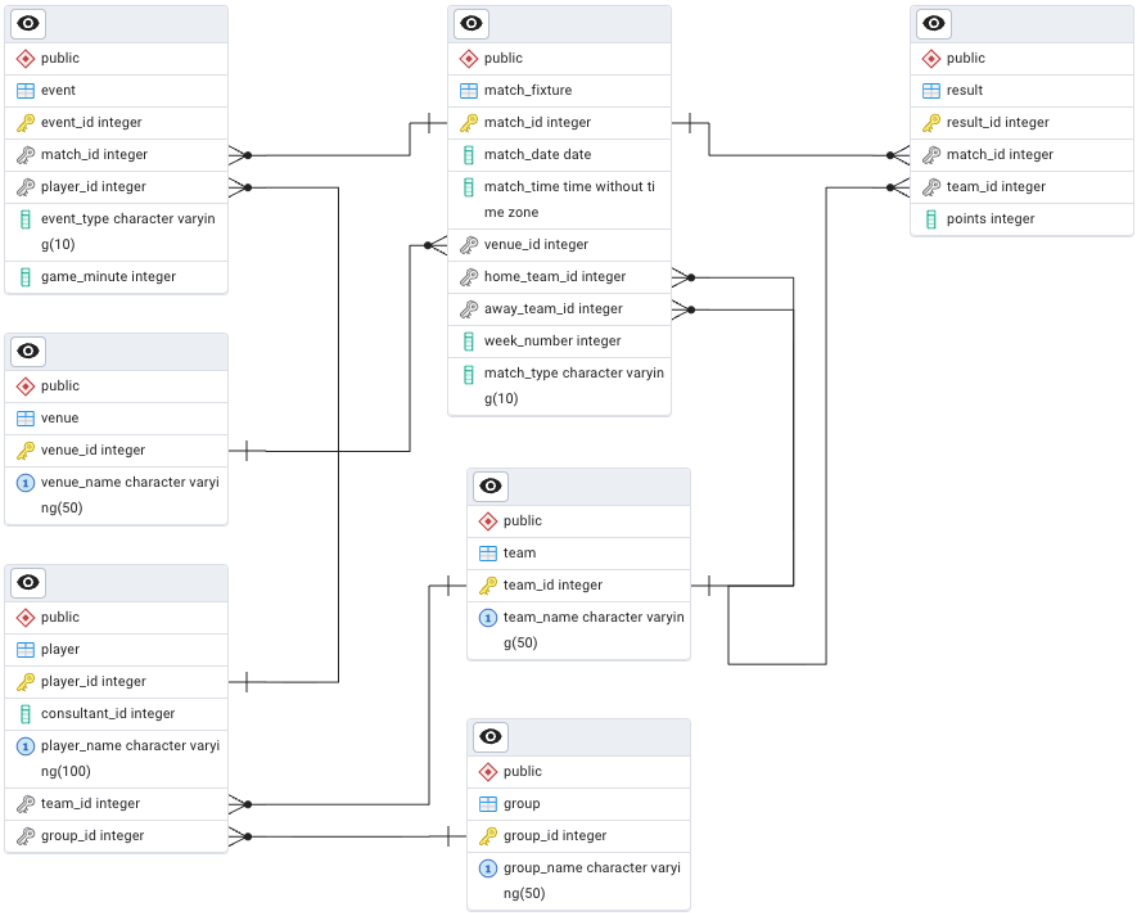

Figure 30 - ER Diagram on PostgreSQL

Figure 30 reflects the ERD of that in Figure 15. This diagram evidences all primary keys and foreign keys as well as data type.